Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 32. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the training loss.

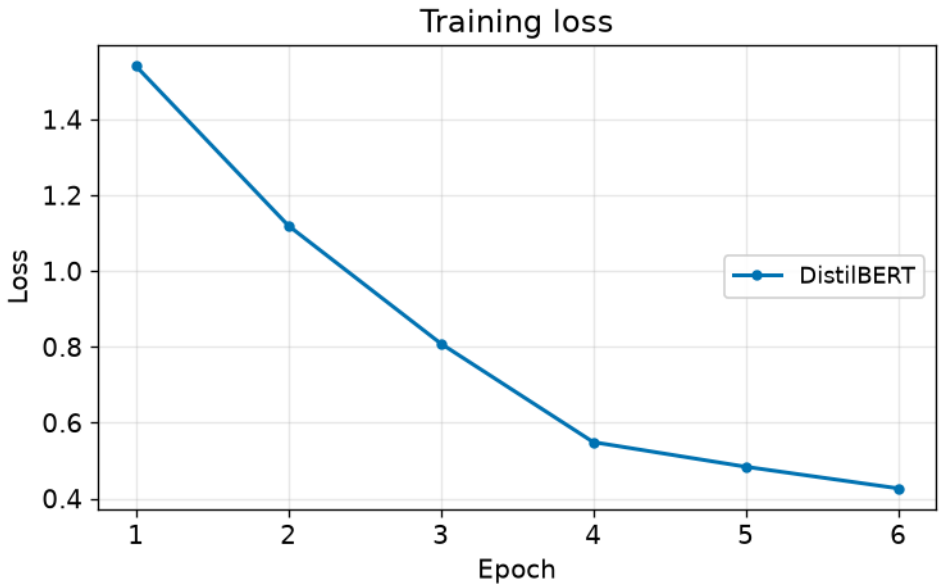


Figure 1: Training loss for the models in this experiment.

As shown in Figure 1, DistilBERT has a loss of 0.423 at epoch 5. DistilBERT has a loss of 0.997 at epoch 2.

Figure 2 shows the accuracy during training.

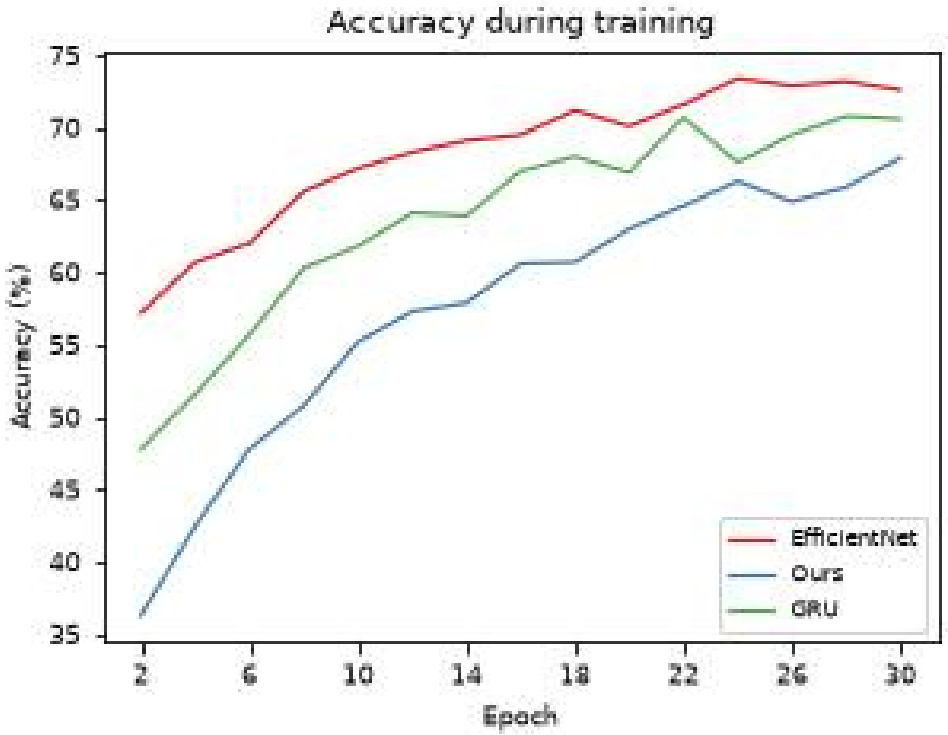


Figure 2: Accuracy during training for the models in this experiment.

At epoch 28, EfficientNet achieves 73.2% accuracy. From epoch 22 to epoch 28, the accuracy of GRU

improves by a relative 5.9%.

Figure 3 shows the accuracy by dataset.

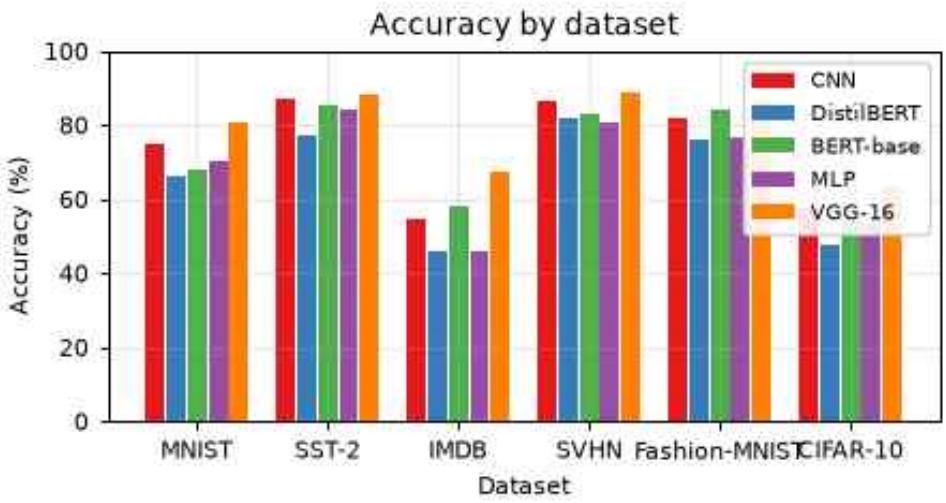


Figure 3: Accuracy by dataset for the models in this experiment.

MLP achieves 56.1% accuracy on IMDB. On MNIST, CNN obtains a accuracy of 125.1%. On MNIST, VGG-16 is 10.2 points better than MLP.

Figure 4 shows the validation accuracy during training.

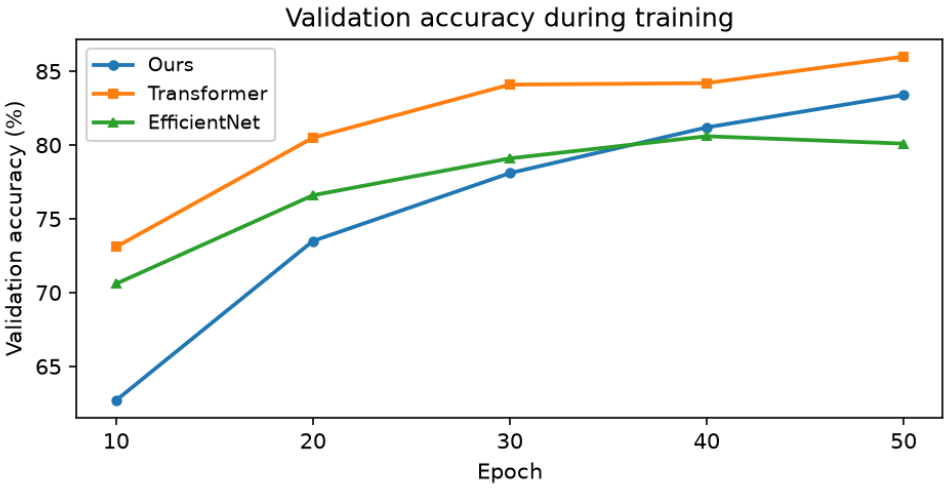


Figure 4: Validation accuracy during training for the models in this experiment.

At epoch 10, Transformer is 10.4 points better than Ours. Transformer reaches 86.0% validation accuracy at epoch 50. As shown in Figure 4, from epoch 30 to epoch 40, the validation accuracy of Transformer improves by a relative 15.4%. At epoch 50, Ours is 16.1 points better than EfficientNet.

3 Discussion

On AG News, Logistic Regression obtains an accuracy of 80.6%. We note that the training was stable for all models.